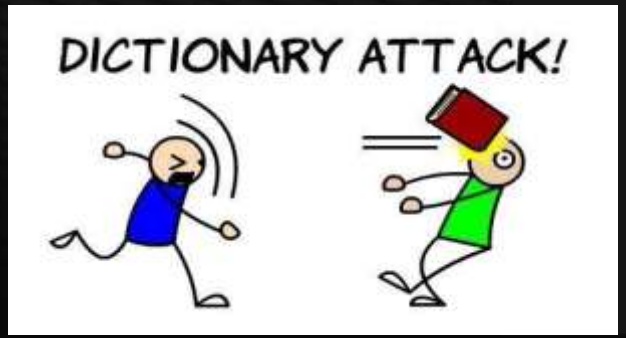
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WHY I'VE TO WASTE MY (PRECIOUS) TIME ON CRYPTOGRAPHY?




```
There are only 10 types
of people in the world:
Those who understand binary,
and those who don't
```

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Crypto is exciting

New attacks!

> Disputes about security!

Improved attacks!

> Proposed fixes!

Even better attacks!

> Emergency upgrades!

Different attacks!

> New protocol versions!

Continual excitement

Tons of research papers

More jobs for cryptographers

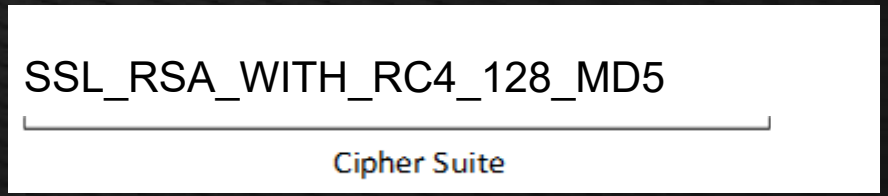
Daniel J. Bernstein (2015)

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Crypto is becoming more and more complex

1996> SSLv3



2018> TLSv1.3

CipherSuite:	TLS_AES_256_GCM_SHA384
Signature:	ecdsa_secp256r1_sha256
Key Exchange:	Groups: X25519

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An example for all...

1987: Ron Rivest designs RC4

Does not publish it

1994: Netscape introduces SSL (Secure Sockets Layer)

SSL supports many options, RC4 is the fastest cipher in SSL

1995: Roos shows that RC4 fails a basic definition of cipher security

(Roos's biases and key reconstruction from permutation)

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More Cryptanalysis

1995	Wagner
1997	Golic
1998	Knudsen–Meier–Preneel–Rijmen–Verdoolaege
2000	Golic
2000	Fluhrer–McGrew
2001	Mantin–Shamir
2001	Fluhrer–Mantin–Shamir
2001	Stubblefield–Ioannidis–Rubin

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1997: IEEE standardizes WEP for 802.11 wireless networks

WEP uses RC4 for encryption

1999: TLS (Transport Layer Security), new version of SSL

RC4 is fastest cipher in TLS

TLS still supports "export keys"

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The blessing of the Father

2001: Rivest response: RC4 is safe in TLS

"Applications which pre-process the encryption key and IV by using hashing and/or which discard the first 256 bytes of pseudo-random output should be considered secure from the proposed attacks. . . .

The 'heart' of RC4 is its exceptionally simple and extremely Efficient pseudo-random generator. . . .
RC4 is likely to remain the algorithm of choice for many applications and embedded systems."

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Even More Cryptanalysis

2002 Hulton
2002 Mironov
2002 Pudovkina
2003 Bittau
2003 Pudovkina
2004 Paul–Preneel
2004 KoreK
2004 Devine
2005 Maximov
2005 Mantin
2005 d'Otreppe

2006 Klein
2006 Doroshenko–Ryabko
2006 Chaabouni
2007 Paul–Maitra–Srivastava
2007 Paul–Rathi–Maitra
2007 Paul–Maitra
2007 Vaudenay–Vuagnoux
2007 Tews–Weinmann–Pyshkin
2007 Tomasevic-Bojanic-Nieto-Taladriz
2007 Maitra–Paul

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... and More Cryptanalysis

2008 Basu–Ganguly–Maitra–Paul
2008 Biham–Carmeli
2008 Golic–Morgari
2008 Maximov–Khovratovich
2008 Akgun–Kavak–Demirci
2008 Beck–Tews
2009 Basu–Maitra–Paul–Talukdar
2010 Sepehrdad-Vaudenay-Vuagnoux
2011 Maitra–Paul–Sen Gupta
2011 Sen Gupta–Maitra–Paul–Sarkar
2011 Paul–Maitra book
2013 Lv–Zhang–Lin
2013 Lv–Lin
2013 Sen Gupta–Maitra–Meier–Paul–Sarkar
2013 Sarkar–Sen Gupta–Paul–Maitra
2013 Isobe–Ohigashi–Watanabe–Morii
2013 AlFardan–Bernstein–Paterson–Poettering–Schuldt
2014 Paterson–Strefler
2015 Sepehrdad–Susil–Vaudenay–Vuagnoux

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2015 Mantin “Bar Mitzvah attack”

2015 Garman–Paterson–van der Merwe “RC4 must die”

2015 Vanhoef–Piessens “RC4 no more”

IETF RFC 7465 “RC4 die die die” prohibits RC4 in TLS

2015.09: Google, Microsoft, Mozilla announce agreement to turn off RC4 in subsequent browser updates…

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Another Example: Elliptic Curves

Introduced in TLS1.1 extension RFC4492

Symmetric Key Size (bits)	RSA and Diffie-Hellman Key Size (bits)	Elliptic Curve Key Size (bits)
80	1024	160
112	2048	224
128	3072	256
192	7680	384
256	15360	521

Table 1: NIST Recommended Key Sizes

ECDSA

ECDH

But what curves can be used?

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Elliptic Curves?

Appendix A. Equivalent Curves (Informative)

All of the NIST curves [11] and several of the ANSI curves [7] are equivalent to curves listed in Section 5.1.1.

Curve names chosen by different standards organizations		
SECG	ANSI X9.62	NIST
sect163k1		NIST K-163
sect163r1		
sect163r2		NIST B-163
...		
sect233k1		NIST K-233
sect233r1		NIST B-233
sect239k1		
sect283k1		NIST K-283
sect283r1		NIST B-283
sect409k1		NIST K-409
sect409r1		NIST B-409
sect571k1		NIST K-571
sect571r1		NIST B-571
...		
secp192r1	prime192v1	NIST P-192
secp224k1		
secp224r1		NIST P-224
secp256k1		
secp256r1	prime256v1	NIST P-256
secp384r1		NIST P-384
secp521r1		NIST P-521

https://safecurves.cr.yp.to/

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Elliptic Curves :-(

File Modifica Visualizza Cronologia Segnalibri Strumenti Aiuto

elliptic curves - Should we trust

https://crypto.stackexchange.com/questions/1026

Cerca

Più visitati Google HotMail Libero ODA Compiler Explorer

Should we trust the NIST-recommended ECC parameters?

127

50

Recent articles in the media, based upon Snowden documents, have suggested that the NSA has actively tried to enable surveillance by embedding weaknesses in commercially-deployed technology -- including at least one NIST standard.

The NIST FIPS 186-3 standard provides recommended parameters for curves that can be used for elliptic curve cryptography. These recommended parameters are widely used; it is widely presumed that they are a reasonable choice.

My question. Can we trust these parameters? Is there any way to verify that they were generated in an honest way, in a way that makes it unlikely they contain backdoors?

Reasons for concern. Bruce Schneier has written that he has seen a bunch of secret Snowden documents, and after seeing them, he recommends classical integer discrete log-based cryptosystems over elliptic curve cryptography. When asked to elaborate on why he thinks we should avoid elliptic-curve cryptography, he writes:

> I no longer trust the constants. I believe the NSA has manipulated them through their relationships with industry.

asked 4 years, 7 mo

viewed 39,7

active

Linked

10

14

6



TLS 1.3 (RFC 8446)

10 Aug 2018

> **No more RSA for key exchange**

> **Ephemeral Diffie-Hellman (ECDHE)** is the only key exchange mechanism **(Perfect Forward Secrecy)**

> Introduced **Curve X25519** (0x001D) and **Curve X448** (0x001E)

> **1-RTT approach** (Entire handhake signed)

> **Redesigned Ciphersuite concept (HKDF and AEAD only)**

> Authentication negotiated using the **signature_algorithms extension** **EdDSA (Curve Ed25519)**, **ECDSA** or **RSA-PSS**

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QUANTUM IS
COMING!!!

Shor's Factoring Algorithm

1994



Peter Shor showed that factoring a number into its primitives is efficient on a quantum computer...

"Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer"

https://arxiv.org/abs/quant-ph/9508027

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Post-Quantum Cryptography

Table 1. Modern cryptosystems and their vulnerability to quantum algorithms.

Cryptosystem	Impact	Comment
RSA	Broken	Shor's algorithm describes an exponential speedup for solving classically difficult number theory problems, such as factoring large prime numbers and solving discrete logarithms.
Diffie-Hellman	Broken	
Elliptical curve	Broken	
Code-based	Not yet broken	These cryptosystems were introduced in the late 1970s, and their security has been well studied. They are not known to be vulnerable to quantum computing (QC) advancements.
Hash-based	Not yet broken	
Lattice-based	Not yet broken	These cryptosystems were introduced in the late 1990s and are believed to be secure against QC advancements.
Multivariate	Not yet broken	
One-time pad (OTP)	Proven unbreakable	Claude Shannon proved the OTP to have perfect secrecy, meaning it is not vulnerable to advancements in QC. Although immune to cryptanalysis, stringent keying requirements limit the OTP's implementation.

https://pqcrypto.org/

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Shor's Factoring Performance (Theoretical)

2012: Fowler presented a way to break 2048bit RSA keys using 1 Billion qubits

2017: O'Gorman optimized the amount to 230 Million qubits (88 hours)

2019: Gheorghiu improved optimizations to 170 Million qubits (24 hours)

2019: Gidney & Ekerå got an historical benchmark with 20 Million qubits (~8 hours)

2025: Gidney (Google) slowed execution time requires 1 Million qubits (~5-7 days)

2026: QLDPC (Quantum Low-Density Parity-Check) codes require 100 K qubits (~hours/days)

March 2026: Caltech Atom Array model potentially requires 10/20 K qubits (~5 days)

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Superconducting Architectures Roadmap

← IBM Tech Atlas

Quantum Roadmap

Strategic milestones

The future of computing is quantum-centric

Client value

2026

Enable the first examples of quantum advantage

Using IBM quantum hardware and HPC, the community will demonstrate the first examples of quantum advantage. Our Nighthawk processor will run circuits with 7,500 gates in up to three 120-qubit modules. To prepare for the future, we will prototype a real-time error correction decoder and demonstrate in the Kookaburra processor a single module comprising a logical processing unit plus a quantum memory.

2028

Diversify quantum advantage and entangle fault-tolerant modules

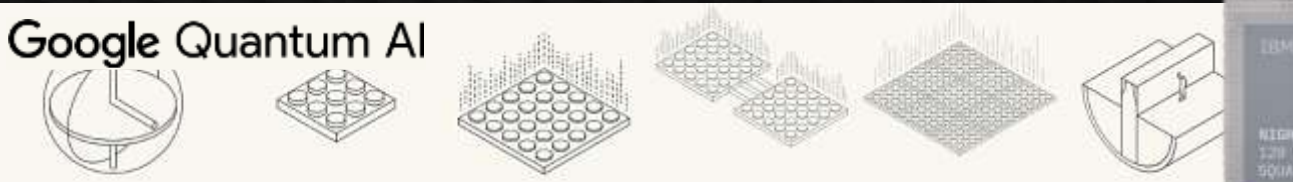
The performance of Nighthawk will improve to allow circuits with up to 15,000 gates on up to 1080 qubits. We will introduce quantum-classical workflow accelerators and computational libraries. We will prototype a complete instruction set architecture for fault-tolerant quantum computing and demonstrate multiple modules and magic state distillation.

2030

Deliver large-scale fault-tolerant quantum computers

The first fault-tolerant quantum computer, Starling, will be available in 2029 with 200 qubits running 100 million gates. By 2033+, we will deliver a 2000 qubits system, called Blue Jay, c… We will scale beyo… computin… and appli…

Google Quantum AI



MILESTONE 1	MILESTONE 2	MILESTONE 3	MILESTONE 4	MILESTONE 5	MILESTONE 6
Beyond classical	Quantum error correction	Building a long-lived logical qubit	Creating a logical gate	Engineering scale up	Large error-corrected quantum computer
Physical Qubits: 54 Logical Qubit Error Rate: -	Physical Qubits: 10^2 Logical Qubit Error Rate: 10^{-2}	Physical Qubits: 10^2 Logical Qubit Error Rate: 10^{-6}	Physical Qubits: 10^4 Logical Qubit Error Rate: 10^{-6}	Physical Qubits: 10^6 Logical Qubit Error Rate: 10^{-6}	Physical Qubits: 10^6 Logical Qubit Error Rate: 10^{-12}
2019	2023				



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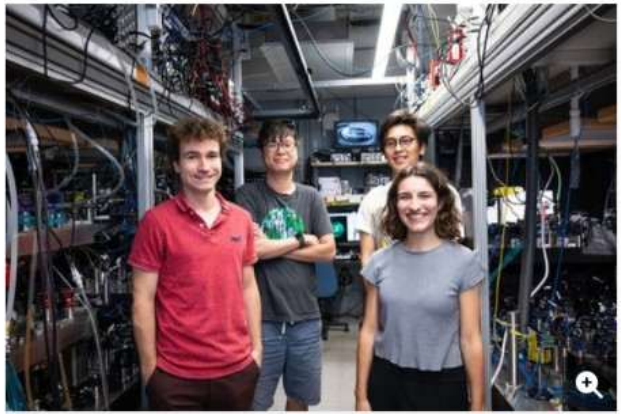
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Neutral-Atom Quantum Computers

Caltech Team Sets Record with 6,100-Qubit Array

September 24, 2025

Quantum computers will need large numbers of qubits to tackle challenging problems in physics, chemistry, and beyond. Unlike classical bits, qubits can exist in two states at once—a phenomenon called superposition. This quirk of quantum physics gives quantum computers the potential to perform certain complex calculations better than their classical counterparts, but it also means the qubits are fragile. To compensate, researchers are building quantum computers with extra, redundant qubits to correct any errors. That is why robust quantum computers will require hundreds of thousands of qubits.



Elie Bataille, Kon H. Leung, Gyohei Nomura, and Hannah Manetsch in a lab at Caltech.

Credit: Caltech/Lance Hayashida

Now, in a step toward this vision, Caltech physicists have created the largest qubit array ever assembled: 6,100 neutral-atom qubits trapped in a grid by lasers. Previous arrays of this kind



THE SHIP HAS SAILED

The NIST Post-Quantum Crypto "Competition"

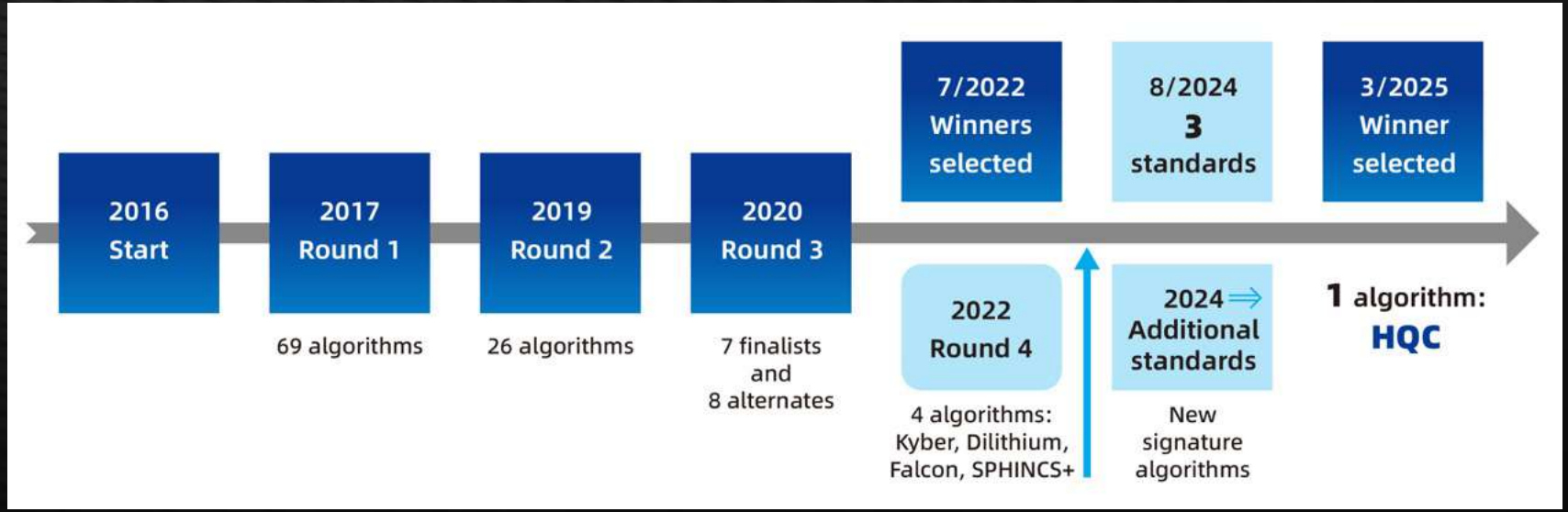
Dustin Moody, NIST

NIST
National Institute of
Standards and Technology
U.S. Department of Commerce

https://csrc.nist.gov/CSRC/media//Projects/Post-Quantum-Cryptography/documents/asiacrypt-2017-moody-pqc.pdf

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NIST PQC Standardization Timeline



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First Set of NIST Published Standards

Algorithm (FIPS Standard)	Function	Cryptographic Family	Legacy Replaced	Public Key Size	Signature / Ciphertext Size	Speed	Selection Year
ML-KEM-768 (FIPS 203)	Encryption (KEM)	Module-Lattice	ECDH	1.184 Bytes	1,088 Bytes (Ciphertext)	Excellent (Microseconds)	2024
ML-DSA-65 (FIPS 204)	Digital Signature	Module-Lattice	ECDSA EdDSA	1.952 Bytes	3,300 Bytes (Signature)	Very Fast	2024
SLH-DSA-192s (FIPS 205)	Digital Signature	Stateless Hash-based	ECDSA	48 Bytes	16,224 Bytes (Signature)	Slow (Milliseconds)	2024
FN-DSA-1024 (FIPS 206)*	Digital Signature	NTRU-Lattice	ECDSA	1.793 Bytes	1,280 Bytes (Signature)	Moderate / Asymmetric	2026 (planned)
HQC-192 (FIPS 207)*	Encryption (KEM)	Code-based	ECDH	4.486 Bytes	4,434 Bytes (Ciphertext)	Moderate	2027 (planned)

Comparison table based on NIST Level 3 Security (AES-192)

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Additional PQC Digital Signatures (3rd Round)

Multivariate	MPC in-the-head		Lattice	Symmetric	Isogeny
UOV	*MinRank*	*MQ*			
Mayo	SDitH	MQOM	HAWK	FAEST	SQIsign
QR-UOV					
SNOVA					
UOV					

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Additional PQC Digital Signatures (3rd Round)

Multivariate	MPC in-the-head	Lattice	Symmetric	Isogeny
~~UOV~~	~~MiRitPank~~	HAWK	FAEST	SQIsign
	~~MQ~~			

ANTHROP\C

Research Policy Commitments Learn News

Frontier Red Team

Discovering cryptographic weaknesses with Claude

Jul 28, 2026

Summary

Using Claude Mythos Preview, researchers at Anthropic have disco improved ways to attack cryptographic algorithms (the mathematic used to keep online data private). The first attack significantly weak digital signature scheme that was built for a post-quantum world. The second identifies a new way to attack round-reduced AES, the most widely used symmetric cipher. These are substantial research advances, but they do not currently affect any production systems. This post describes both findings in more detail and discusses the implications for cryptography in an age of powerful AI models.

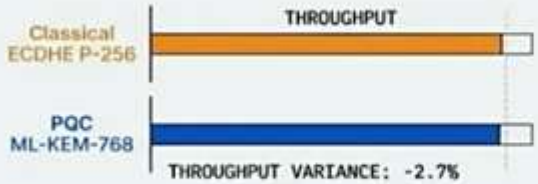
On July 28, 2026, Anthropic announced that it used an AI model to discover a vulnerability in a lattice-based digital signature algorithm called HAWK, which was under consideration for standardization. The HAWK development team subsequently withdrew its algorithm from consideration, and it will not be standardized or deployed. This finding does not affect any of NIST's finalized PQC standards, such as ML-KEM and ML-DSA, which rely on different mathematical foundations that remain secure, and which are ready for implementation now.

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ML-KEM across Network Architectures

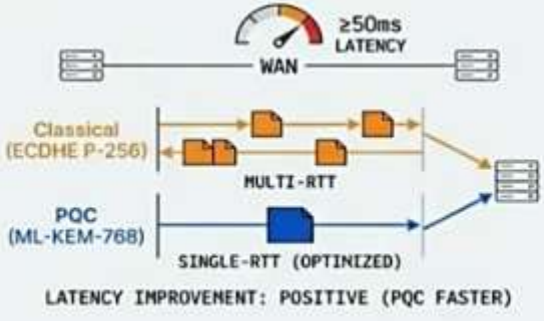
Compute is Parity

ML-KEM-768 math does not slow down modern servers. Cryptographic encapsulation throughput is within **2.7%** of classical ECDH.



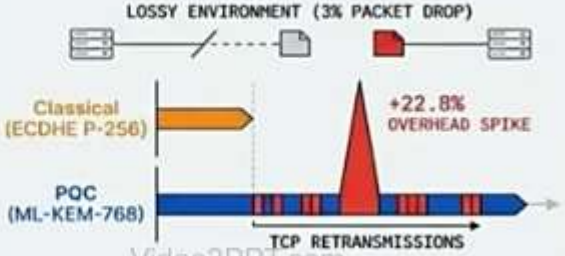
Long-Haul is Faster

Over high-latency WANs (≥50ms), PQC's one-round-trip design actually outperforms classical key exchanges.



Loss is the Enemy

In lossy environments (3% packet drop), PQC handshake overhead spikes up to **+22.8%** due to TCP retransmissions of massive ciphertexts.

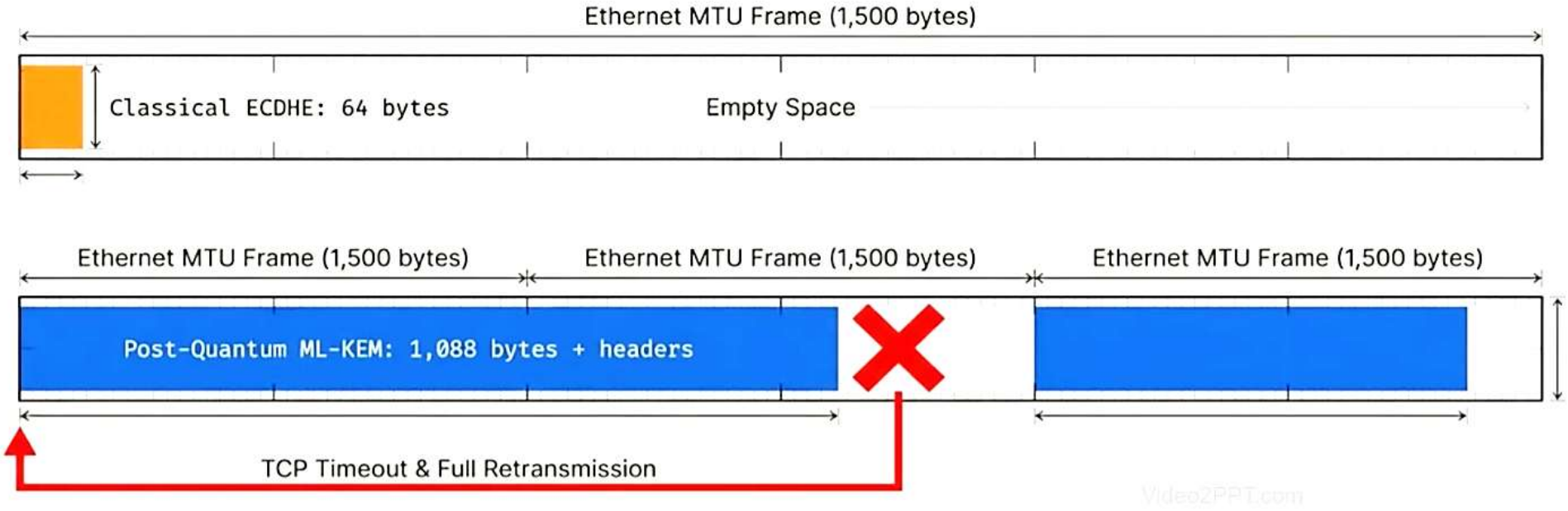


Video2PPT.com

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ML-KEM Fragmentation Problem



Video2PPT.com

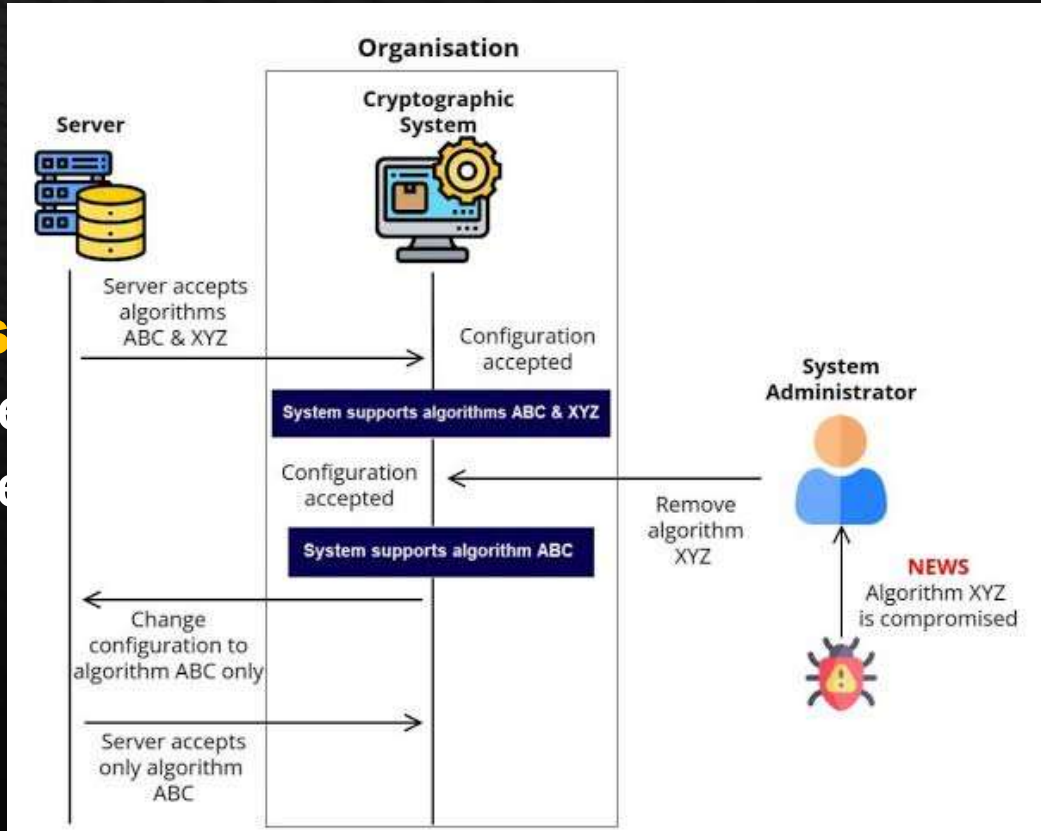
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The new Crypto-Buzzword

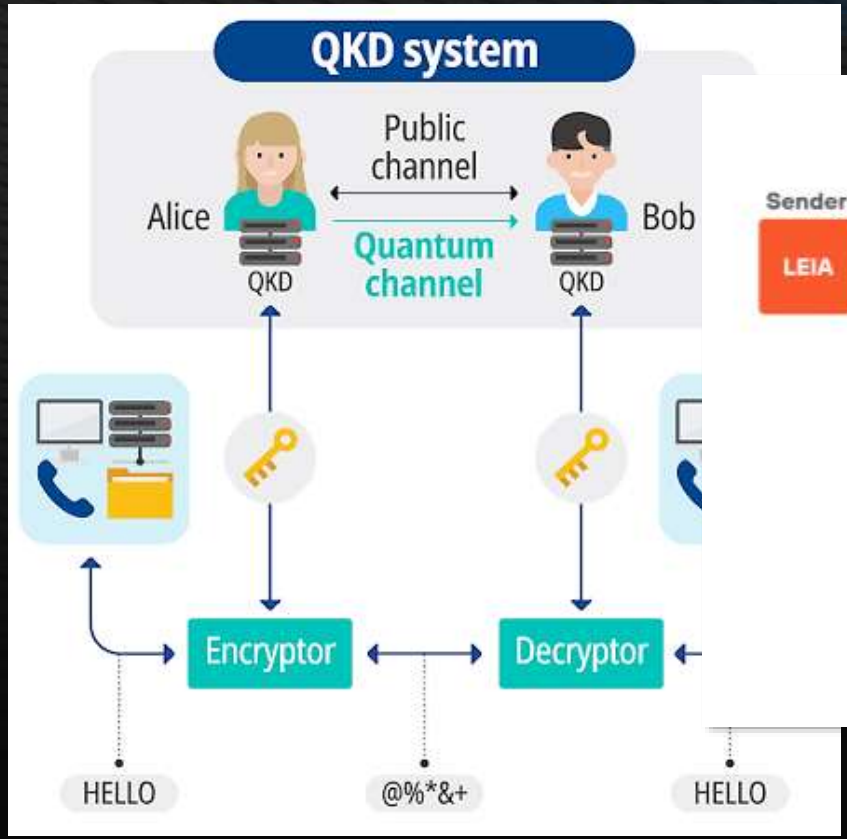
Crypto-agility

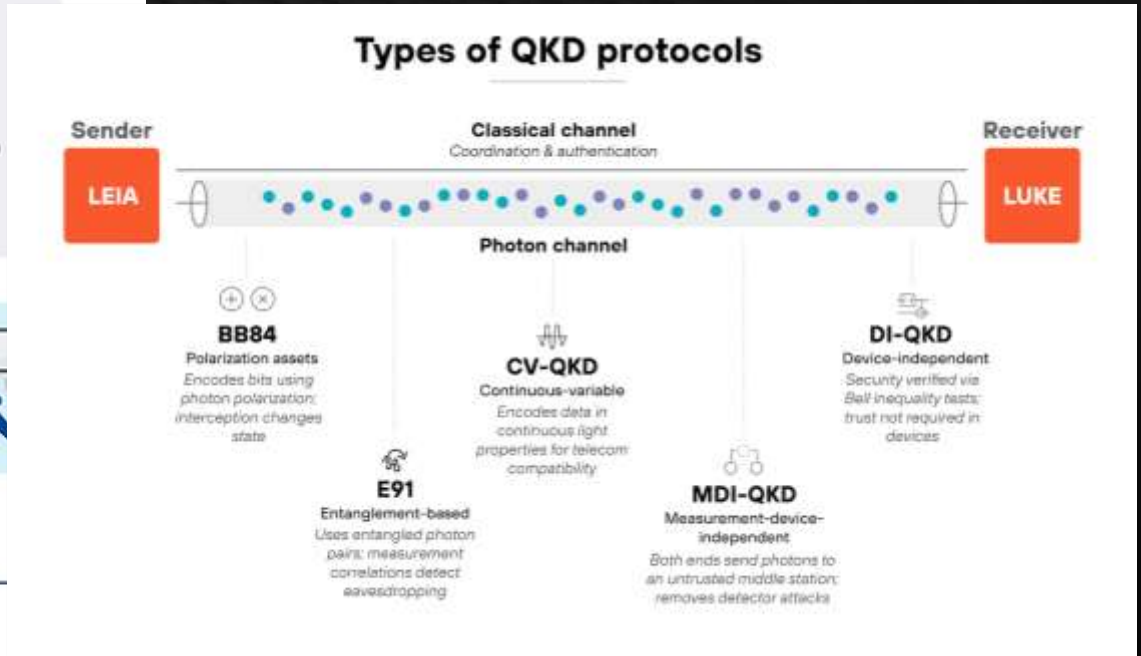
> "the capability to **rapidly**
> **algorithms and protocols**
> without full reviews of the
> modifications of the unde



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Quantum Key Distribution (QKD)





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2023-2027 EuroQCI



AIRBUS CNR-INO INRiM LEONARDO orange pwc TELESPAZIO

Joint Press Release

Un consorzio di player digitali europei per progettare la futura rete Internet quantistica dell'UE

Obiettivo del progetto è assicurare comunicazioni ultrasicure per infrastrutture critiche e istituzioni governative

@AirbusDefence, @EU_Commission, @Leonardo, @Orange, @PWC, @Telespazio, @Consiglio Nazionale delle Ricerche, @Istituto Nazionale di Ricerca Metrologica

Bruxelles, 31 maggio 2021 – La Commissione europea ha selezionato un consorzio di aziende e istituti di ricerca per progettare la futura rete europea di comunicazione quantistica, EuroQCI (*quantum communication infrastructure*), che consentirà la comunicazione ultrasicura tra infrastrutture critiche e istituzioni governative in tutta l'Unione europea.

Del consorzio, guidato da Airbus, fanno parte Leonardo, Orange, PwC France e Maghreb, Telespazio (Leonardo 67%, Thales 33%), il Consiglio Nazionale delle Ricerche (CNR) e l'Istituto Nazionale di Ricerca Metrologica (INRiM).

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Real-World QKD Limitations

#1> Distance & Fiber Attenuation Constraints

Single photons degrade rapidly inside standard optical fibers (80-120 km)

#2> No-cloning Theorem Bottleneck

Quantum states cannot be copied, amplified, or boosted by repeaters

#3> Secret Key Rate (SKR)

Throughput drops drastically as distance increases (Kbits)

#4> Hardware Dependency

Requires proprietary, state-of-the-art expensive hardware

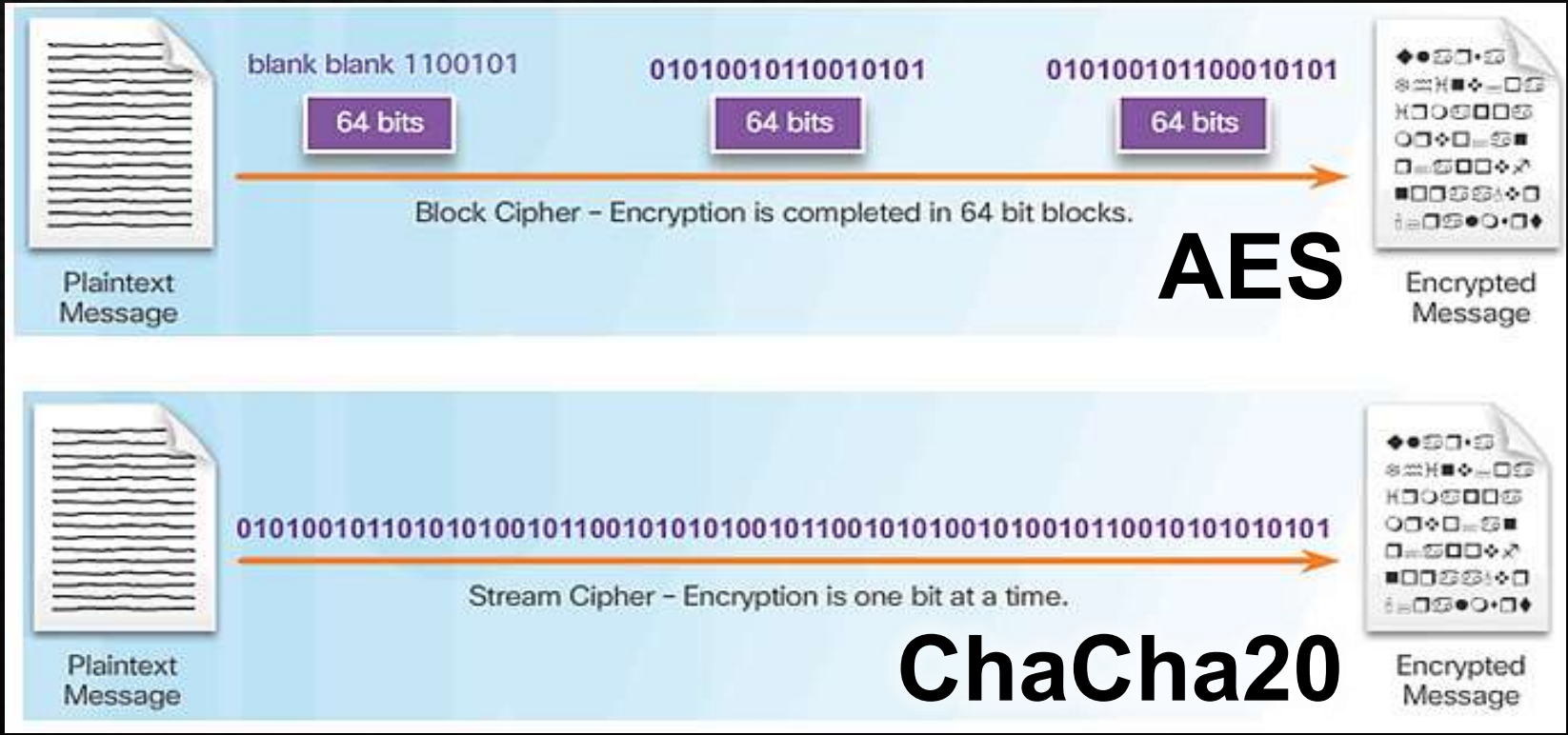
#5> Vulnerable to Denial of Service (DoS)

Attackers can drop the connection bending the fiber or injecting optical noise

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What about Symmetric Encryption?



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QUANTUM IS
COMING!!!

Glover Algorithm

1996



Lov Kumar Grover devised a quantum search algorithm that finds the unique input of a black box function for a specific output value using $O(\sqrt{N})$ evaluations...

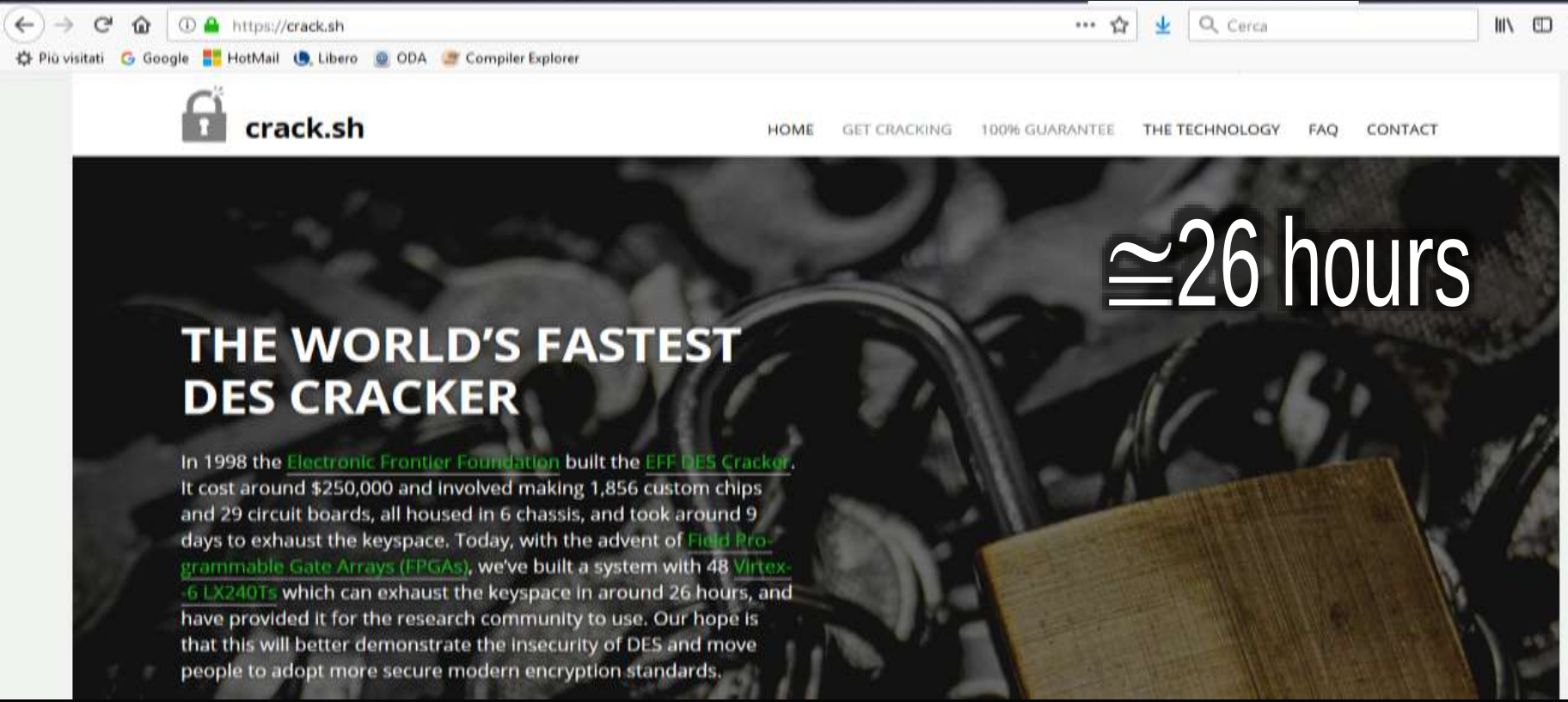
"A fast quantum mechanical algorithm for database search"

https://arxiv.org/abs/arXiv:quant-ph/9605043

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One step back: Brute Force attacks



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45,360 ×
304-core Huawei LX2-ARMv9
Rmax 2,198 petaflops

2,74 EsaFLOP

https://en.wikipedia.org/wiki/TOP500

1000 Flops required per combination check (very optimistic)

Combination / second = 2,74 x 10^{18} Flops / 1000 = 2,74 x 10^{15}

https://www.embedded.com/design/embedded-internet-design/4372428/How-secure-is-AES-against-brute-force-attacks-

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Brute Force using NSCS LineShine

Key Size	Time to Crack
DES (56 bit)	13 seconds
AES (128 bit)	1.97×10^{15} years
AES (256 bit)	6.70×10^{53} years

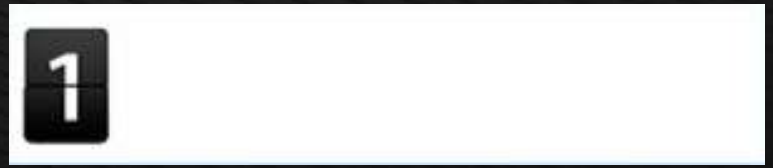
The current estimated age of the universe is **13.8 billion years** (13.8×10^{9})

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Brute Force Physics Constraints

Brute Force == Bit Flipping



Can we define a physical model for a Brute Force activity?

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Brute Force Physics Constraints

Second law of Thermodynamics:

"Energy is required to do a certain amount of work"

$$E = k \times T = 1.38064852 \times 10^{-16} \times T$$

Background radiation in outer space is about 2.72548 Kelvin

$$E = 1.38 \times 10^{-16} \times 2.72548 = 3.762 \times 10^{-16} \text{ ergs}$$

https://pthree.org/2016/06/19/the-physics-of-brute-force/

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Brute Force Physics Constraints



Total annual energy output of the Sun is about 1.2×10^{41} ergs

$$1.2 \times 10^{41} / 3.762 \times 10^{-16} = 3.19 \times 10^{56} \text{ bit} \cong 2^{187}$$



Total annual energy output of a Supernova is about 10^{51} ergs

$$10^{51} / 3.762 \times 10^{-16} = 2.66 \times 10^{66} \text{ bit} \cong 2^{220}$$



Total annual energy output of a Hypernova is about 10^{53} ergs

$$10^{53} / 3.762 \times 10^{-16} = 2.66 \times 10^{68} \text{ bit} \cong 2^{227}$$

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Symmetric Key Lenght (Classic Crypto)

Rule #1 > Minimum 128 bit keys

Rule #2> Maximum 256 bit keys

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Glover Algorithm Performance

2015

Applying Grover's algorithm to AES: quantum resource estimates

Markus Grassl[1], Brandon Langenberg[2], Martin Roetteler[3], and Rainer Steinwandt[2]

[1] Universität Erlangen-Nürnberg & Max Planck Institute for the Science of Light, Günther-Scharowsky-Straße 1, Bau 24, 91058 Erlangen, Germany, Markus.Grassl@fau.de

[2] Florida Atlantic Universit

[3] Microsoft Research

	#gates		depth		#qubits
k	T	Clifford	T	overall	
128	$1.19 \cdot 2^{86}$	$1.55 \cdot 2^{86}$	$1.06 \cdot 2^{80}$	$1.16 \cdot 2^{81}$	2,953
192	$1.81 \cdot 2^{118}$	$1.17 \cdot 2^{119}$	$1.21 \cdot 2^{112}$	$1.33 \cdot 2^{113}$	4,449
256	$1.41 \cdot 2^{151}$	$1.83 \cdot 2^{151}$	$1.44 \cdot 2^{144}$	$1.57 \cdot 2^{145}$	6,681

Table 5. Quantum resource estimates for Grover's algorithm to attack AES-k, where $k \in \{128, 192, 256\}$.

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Glover Algorithm Performance

Scenario	Single Operation Time	Total Time (AES-128)	Note
Sci-Fi (Super-fast)	1 nanosecond (10^{-9} s)	~584 years	Physically improbable due to quantum error correction (QEC) overhead
Optimistic (Research Target)	1 microsecond (10^{-6} s)	~584.542 years	Standard target assumed in scientific literature (e.g., NIST/Grassl estimates)
Conservative (Current Tech)	1 millisecond (10^{-3} s)	~584 million years	Closer to speeds of neutral atom or trapped-ion systems without error correction

OK, let's parallelize the search **on multiple Quantum Computers...**

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But Glover is not Shor

With N quantum systems running Grover, you get only $\sqrt{N}$ faster

Christof Zalka*
zalka@t6-serv.lanl.gov

November 26, 2024

Abstract

I show that for any number of oracle lookups up to about $\pi/4\sqrt{N}$, Grover's quantum searching algorithm gives the maximal possible probability of finding the desired element. I explain why this is also true for quantum algorithms which use measurements during the computation. I also show that unfortunately quantum searching cannot be parallelized better than by assigning different parts of the search space to independent quantum computers.

How Break AES-128 in 6 Months...

Scenario	Quantum Cluster Size
Sci-Fi (Super-fast)	1,37 Million computers
Optimistic (Research Target)	1,36 Trillion computers
Conservative (Current Tech)	1,36 Quintillion computers

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Symmetric Key Lenght (Post Quantum Crypto)

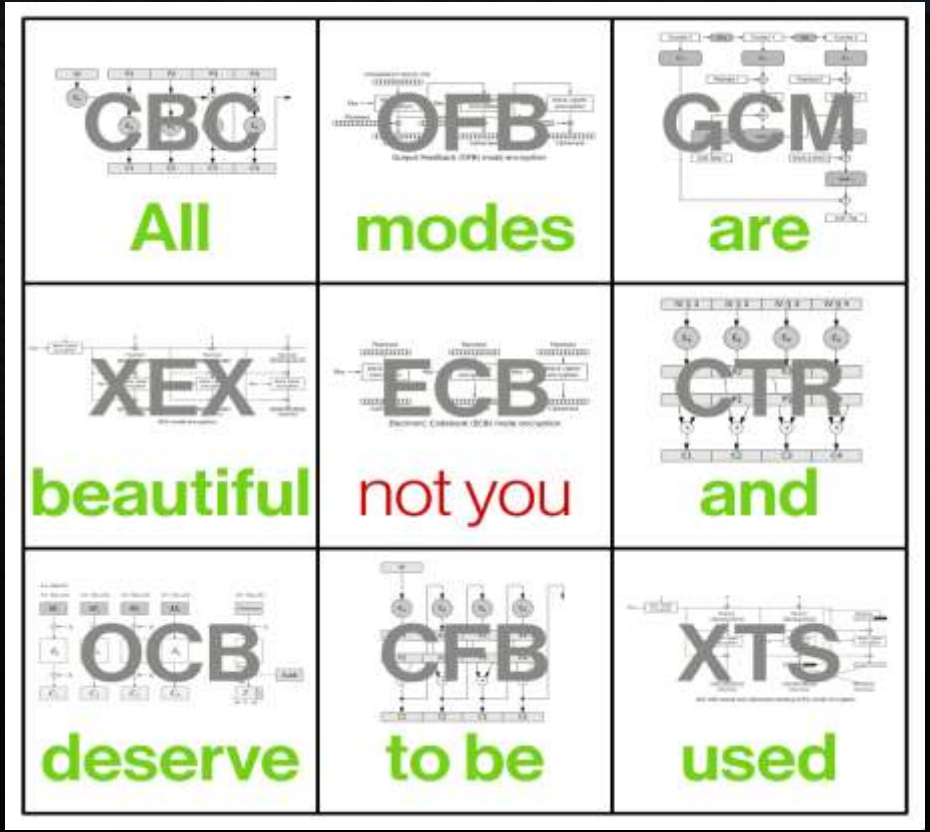
Rule #1> Use 256 bit keys

Rule #2 > Use 128 bit keys for ephemereal data

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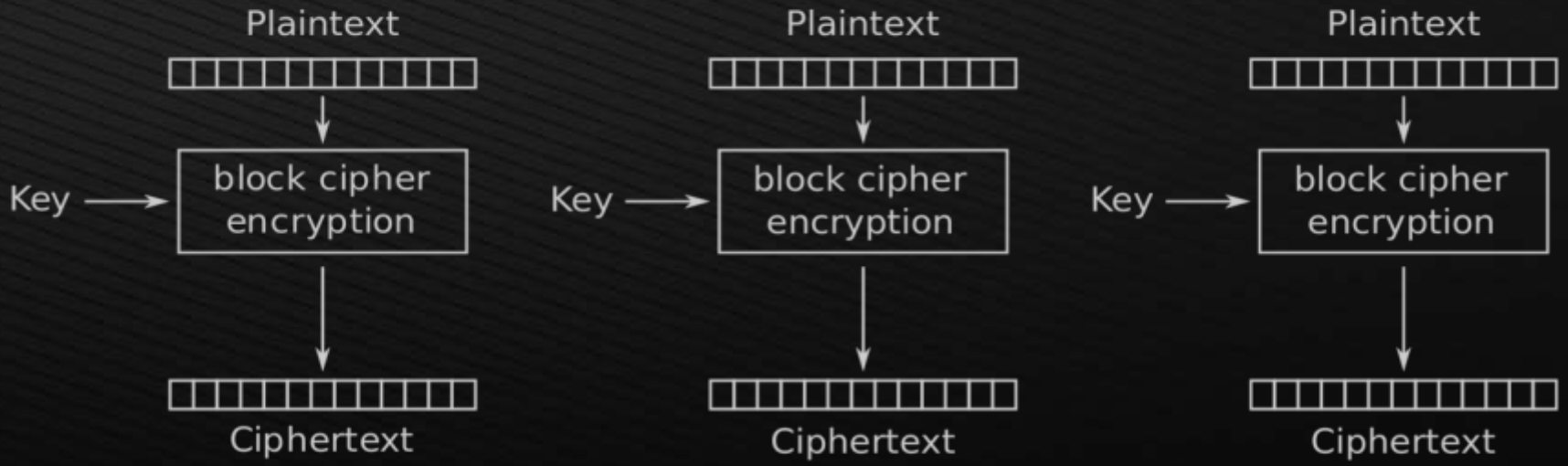
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The Real Issue: Block Ciphers modes of Operation



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Electronic Codebook (ECB) mode encryption

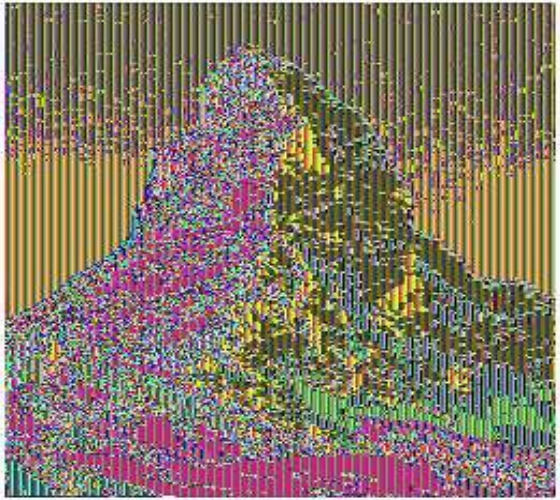
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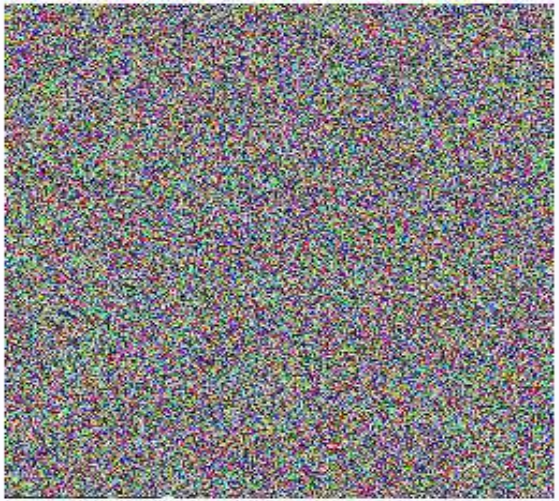
Why ECB is bad...



Original picture



With ECB Block Mode

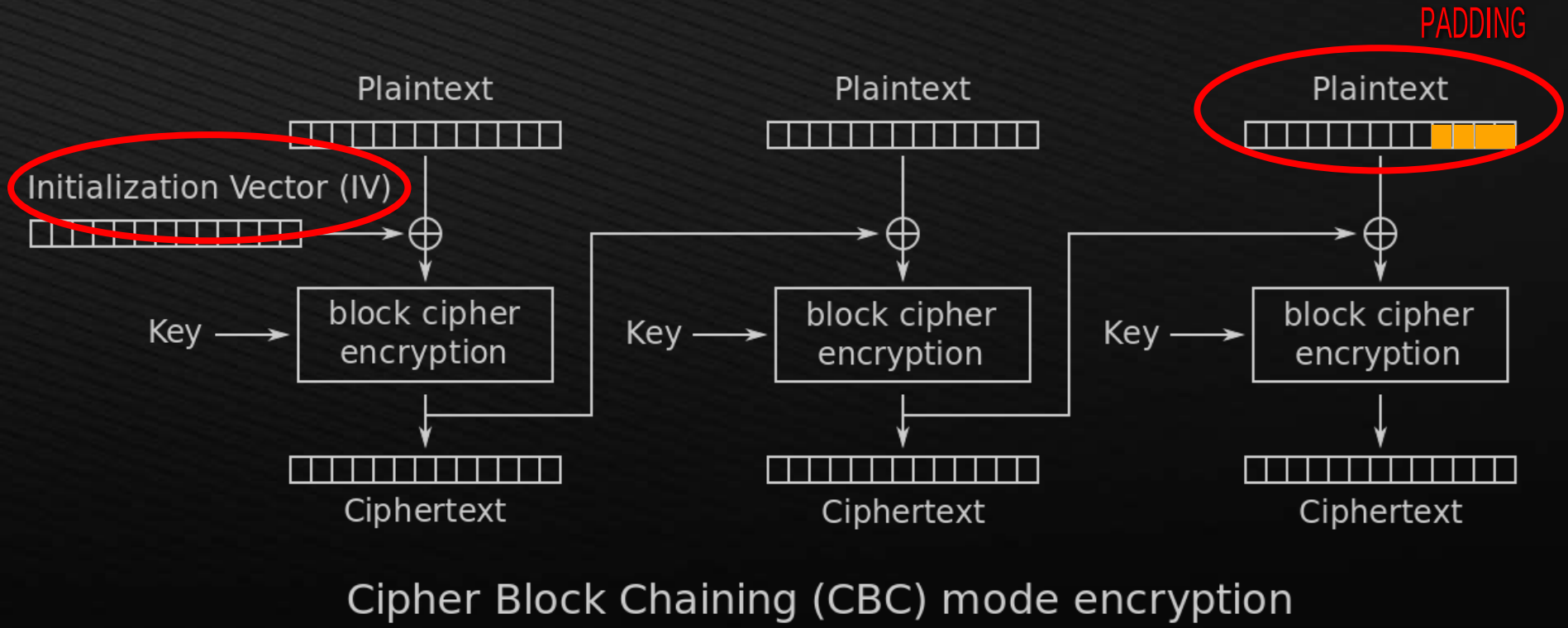


With any other Block Mode

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Why also the others are bad...

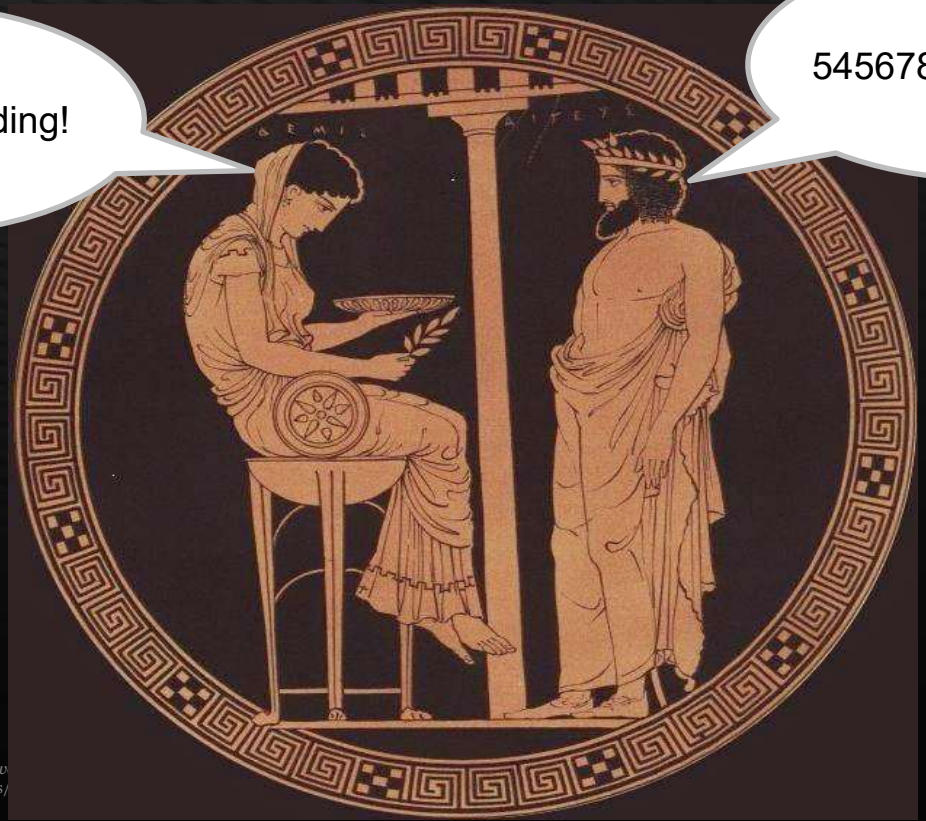


Cipher Block Chaining (CBC) mode encryption

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Padding Oracle Attack (definition)

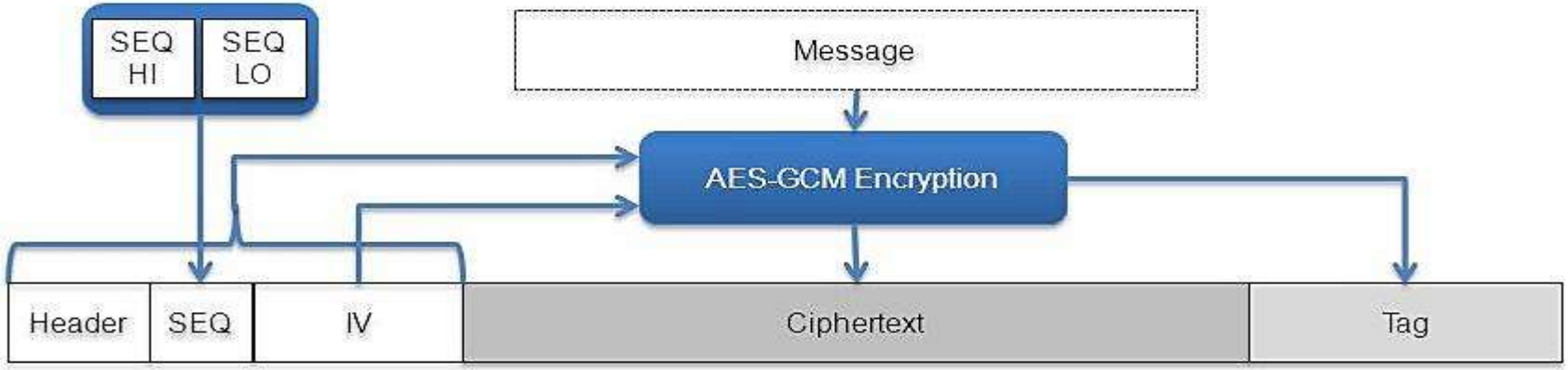


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Authenticated Encryption with Associated Data

Simultaneously provides
confidentiality, integrity, and authenticity assurances



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Why not Stream Ciphers?

Written by Derek Zimmer on 5/13/2014

Google Is Pushing New Cipher Suites - All About ChaCha20 and Poly1305

After the Snowden revelations, tech companies have been scrambling to find newer and stronger encryption schemes to boost confidence in encryption around the world as a way to safeguard your data from intrusion by advanced threats like governments. A lot of companies have moved to 2048-bit asymmetric keys and 256-bit symmetrical ciphers in an effort to strengthen current protocols against brute force attacks with enormous distributed networks like botnets or world-class supercomputers.

There also has been a large outcry for stronger protocols with fewer theoretical attacks against them. The most commonly used ciphers (RC4 and AES-CBC) have had practical attacks executed in the wild that demonstrated vulnerability.

New protocols are in the works such as AES in Galois Counter Mode, the Skein Hash Protocol, and Keccak but they are heavy protocols that require a lot of computing power to handle large amounts of data. They are heavy on devices with traditionally weaker or low power processors like smartphones and tablets.

In walks Dan Bernstein with Chacha20 and Poly1305.

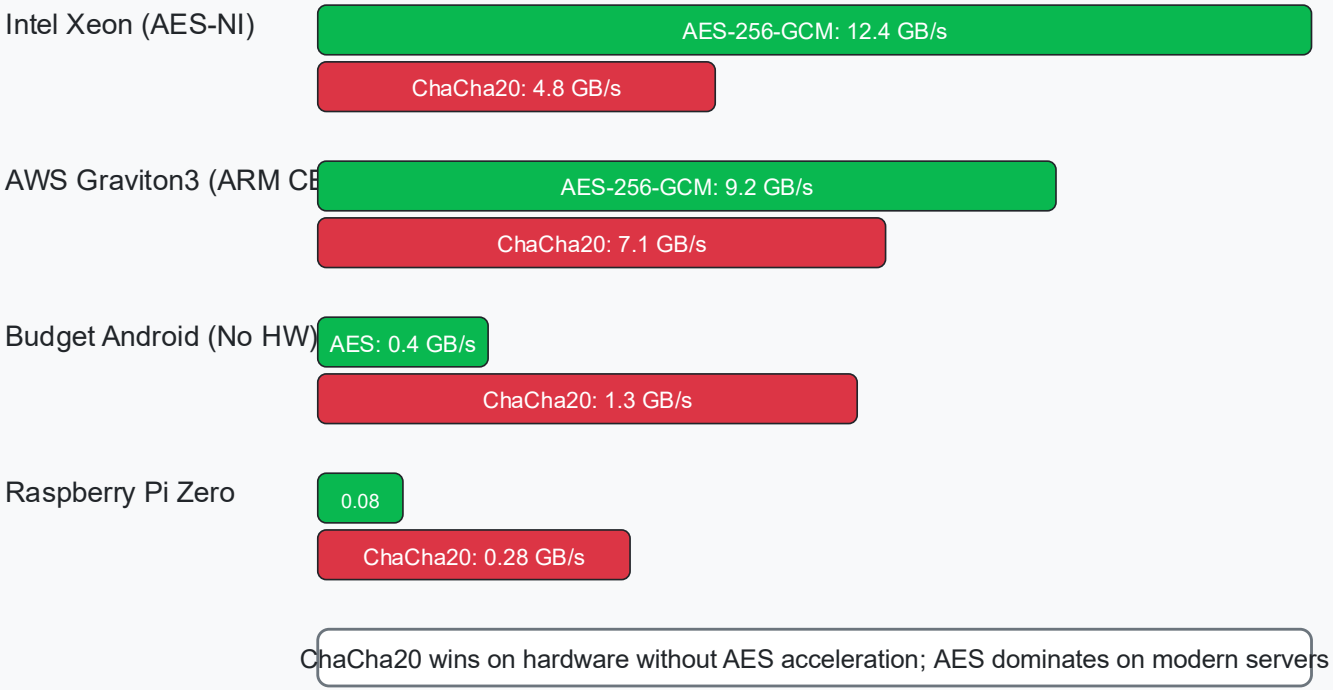
ChaCha20 is a new stream cipher that is functionally similar to the old Salsa20 cipher. Poly1305 is for message authentication and is both less cpu intensive and uses less bandwidth than SHA1/2/3 or Skein, while still allowing for perfect message integrity assurance as long as the cipher it is checking is known to be secure. Together, they offer a substantial performance advantage for devices that may be limited by their processors. ChaCha20+Poly1305 can be as much as 300% faster than AES-256-GCM with SHA-1 authentication.

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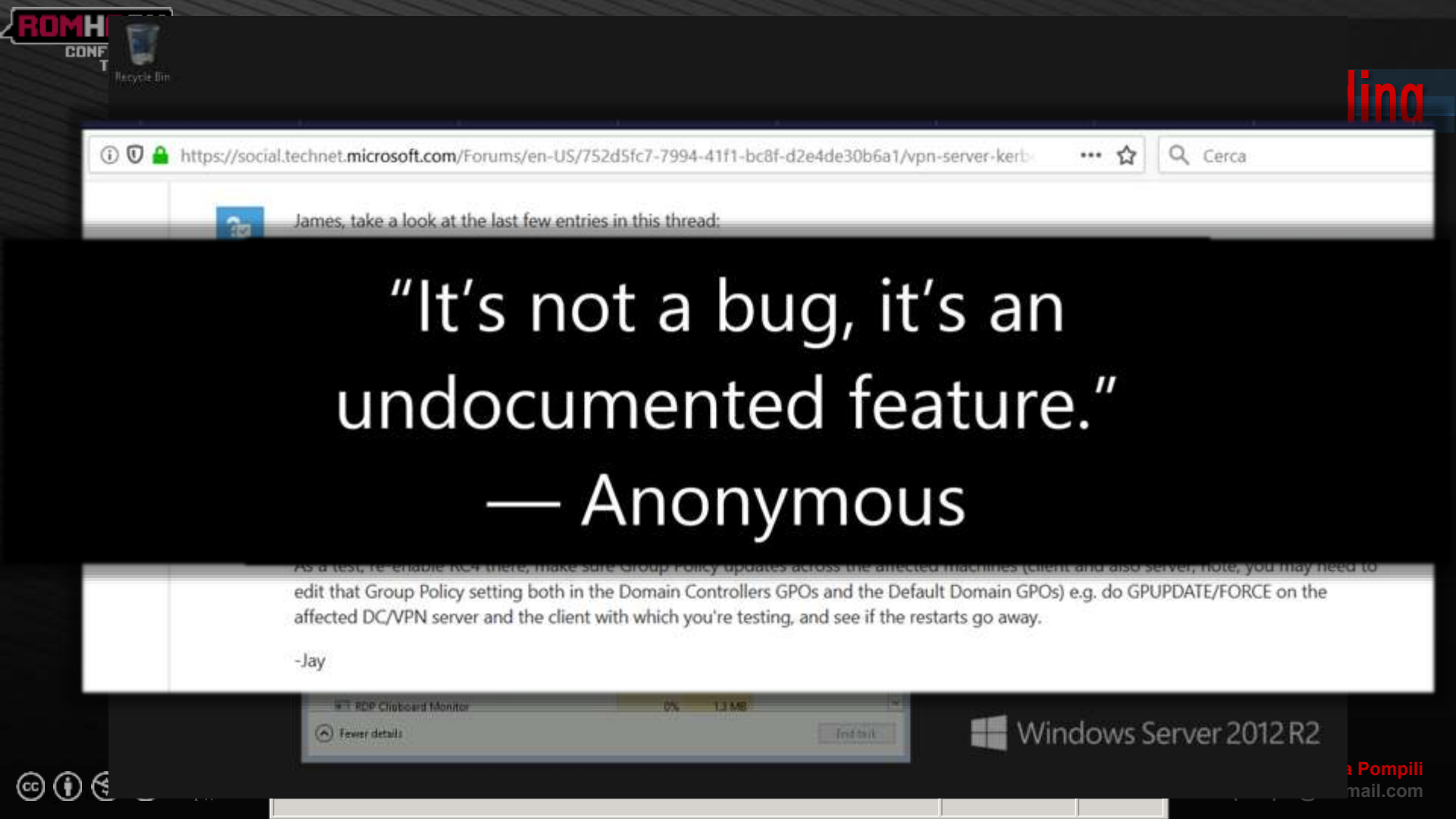
Why not Stream Ciphers?

Throughput Comparison: AES-GCM vs ChaCha20 (GB/s)



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"It's not a bug, it's an undocumented feature."

— Anonymous

ROMHACK
CONFERENCE
TRAINING
CAMP

Questions?
English

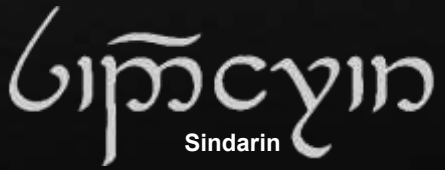
¿Preguntas?
Spanish

مَطَالِب أَيَّة
Arabic

вопросы?
Russian

Domande?
Italian

Ερωτήσεις?
Greek


Sindarin

tupoQghachmey
Klingon

質問
Japanese

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